

Project Proposal: Modeling Simple Decisions and the Speed-Accuracy Trade-off Using the Drift Diffusion Model

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1. Introduction: What is this project about?

When you have to make a quick choice between two options, for example, deciding whether a faint image appeared on the **left** vs **right** of a screen, your brain gathers visual information over a few hundred milliseconds until it feels confident enough to make a choice.

This leads to two common-sense observations:

1. **Easy tasks:** If the image is bright and obvious, you decide very quickly and rarely make mistakes.
2. **Hard tasks:** If the image is faint or blurry, you take longer to decide and make more mistakes.

The classic computational model used to explain this behavior is the **Drift Diffusion Model (DDM)**. It treats decision-making like an evidence counter:

- You start at neutral (0).
- As you look at the screen, sensory evidence pushes your internal counter toward the "left" or "right" threshold
- Because neural signals are noisy, the counter doesn't move in a straight line and instead it moves up and down randomly as it drifts.
- The moment the counter hits either threshold, you commit to that choice and press the button.

In this project, we will build a simulation of this model in Python to study how evidence accumulation explains both **what people/animals choose** and **how long they take to respond**.

2. Research Question*

Our specific question is: **"How does evidence accumulation explain both choice accuracy and response times across easy and difficult tasks? Furthermore, how does adjusting the decision threshold explain the "speed vs. accuracy" trade-off, and what does this assume about how the brain makes decisions?"**

What we plan to do:

1. **Simulate the model in Python:** Write a simple script that simulates trial-by-trial decisions.
2. **Vary task difficulty:** Test whether the model reproduces the expected patterns across different stimulus contrast levels (from 0% contrast up to 100% contrast).
3. **Test the Speed-Accuracy Trade-off:** Change the threshold height to see how being "cautious" (high threshold) versus "in a rush" (low threshold) changes accuracy and reaction time.
4. **Examine the model's assumptions:** Discuss what the model assumes about how the brain processes information and where those assumptions make sense or fall short.

3. How the Model Works

The DDM models decision-making as a continuous, noisy accumulation of evidence $x(t)$ starting at neutral ($x(0) = 0$). At each discrete time step Δt , evidence updates as:

$$x(t + \Delta t) = x(t) + v \cdot \Delta t + \text{noise}$$

- **Drift Rate (v):** Reflects stimulus strength/clarity. Higher contrast produces larger v (faster drift toward target); faint or zero-contrast stimuli yield slow or zero drift ($v \approx 0$).
- **Noise:** Random neural fluctuations modeled as Gaussian noise ($\sigma\sqrt{\Delta t}$).
- **Boundaries ($\pm a$):** Deliberation terminates when $x(t)$ crosses $+a$ (choose Left) or $-a$ (choose Right), yielding the **decision time** (T_d).

The total **reaction time (RT)** incorporates physical delays via non-decision time ($t_0 \approx 0.2$ s), covering both sensory encoding and motor execution (time taken to act):

$$\text{RT} = T_d + t_0$$

4. Model Assumptions

1. The brain accumulates evidence rather than guessing on snapshots.
2. Caution is controlled by boundary separation (a).
3. Caution stays constant throughout the trial.
4. Deliberation and motor execution are separate.

5. Key References

1. **Bogacz, R., Brown, E., Moehlis, J., Holmes, P., & Cohen, J. D.** (2006). The physics of optimal decision making: a formal analysis of models of performance in two-alternative forced-choice tasks. *Psychological Review*, 113(4), 700–765.
2. **International Brain Laboratory.** (2021). A standardized and reproducible method to measure decision-making in mice. *eLife*, 10, e63711.
3. **Ratcliff, R., & McKoon, G.** (2008). The diffusion decision model: theory and data for two-choice decisions. *Neural Computation*, 20(4), 873–922.